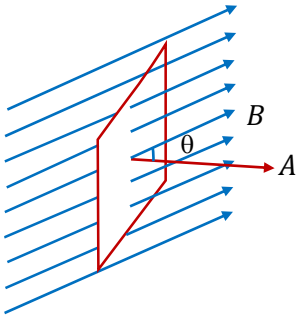
02 Electromagnetic Induction (EMI)

Magnetic Flux

Faraday's great insight lay in discovering a simple mathematical relation to explain the series of experiments he carried out on electromagnetic induction. However, before we state and appreciate his laws, we must get familiar with the notion of magnetic flux, Φ_B. Magnetic flux is defined in the same way as electric flux is defined in the chapter Electrostatics.



A plane of surface area ***A*** placed in a uniform magnetic field ***B***.

Magnetic flux through a plane of area A placed in a uniform magnetic field B can be written as $\phi_B = B.A = BA\cos\theta$

where θ is angle between B and A. The notion of the area as a vector has been discussed earlier in electrostatics. Above Equation can be extended to curved surfaces and nonuniform fields.

$$\phi = \int \vec{B}.d\vec{A}$$

Faraday's laws of electromagnetic induction

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Faraday stated experimental observations in the form of laws called Faraday's laws of electromagnetic induction. The law is stated below.

The magnitude of the induced emf in a circuit is equal to the time rate of change of magnetic flux through the circuit.

Mathematically, the induced emf is given by

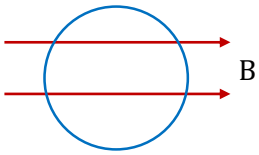
$$E = -\frac{d\phi}{dt}$$

(–) sign indicates that the emf will be induced in such a way that it will oppose the change of flux (Len'z law).

SI unit of magnetic flux = Weber.

Illustration 1

A coil is placed in a constant magnetic field. The magnetic field is parallel to the plane of the coil as shown in figure. Find the emf induced in the coil.



Solution:

ϕ = 0 (always) since area is perpendicular to magnetic field.

$\therefore$ $\text{emf}_{\text{induced}} = 0$

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Illustration 2

Find the emf induced in the coil shown in figure. The magnetic field is perpendicular to the plane of the coil and is constant.

Solution:

$\phi = BA$ (always)

= const.

$\therefore$ $emf_{induced} = 0$

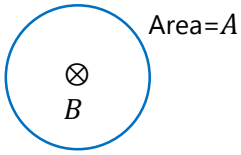
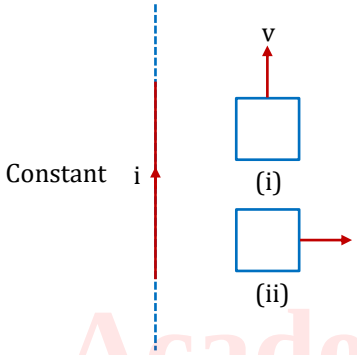


Illustration 3

Figure shows a long current carrying wire and two rectangular loops moving with velocity v. Find the direction of current in each loop.



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Solution:

In loop (i) no emf will be induced because there is no flux change.

In loop (ii) emf will be induced because the coil is moving in a region of decreasing magnetic field which is inward in direction. Therefore, to oppose the flux decrease in inward direction, current will be induced such that its magnetic field will be inwards. For this direction of current should be clockwise.

Lenz's Law (Conservation of Energy Principle)

The current induced in the coil will produce heat. From energy conservation, if heat is produced there must be an equal decrease of energy in some other form, here it is the kinetic energy of the moving magnet. Thus, the magnet must slow down. So, we can justify that the **Lenz's law is conservation of energy principle**.

Motional emf from Lorentz force

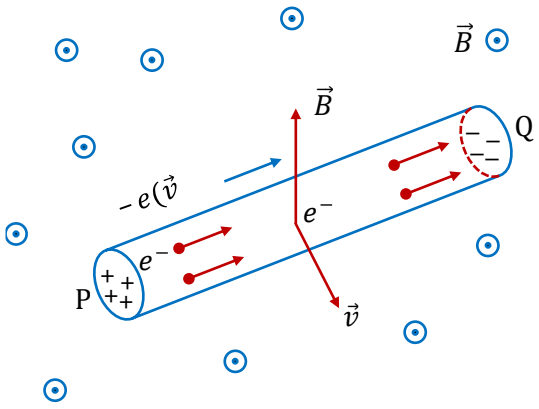
A conductor PQ is placed in a uniform magnetic field B, directed outwards, perpendicular to the plane of the paper. PQ is moved with a velocity v, the free electrons of PQ also move with the same velocity. The electrons experience a magnetic Lorentz force,

$$F = e\vec{v} \times \vec{B}$$

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According to right hand rule, this force acts in the direction PQ and hence the free electrons will move towards Q. Negative charge accumulates at Q and positive charge at P. An electric field E is setup in the conductor from P to Q. Force exerted by electric field on the free electrons is,

$$\vec{F}_e = e\vec{E}$$



The accumulation of charges at the two ends continues till these two forces balance each other.

So $\vec{F}_m = -\vec{F}_e \Rightarrow e(\vec{v}\times\vec{B}) = -e\vec{E} \Rightarrow \vec{E} = -(\vec{v}\times\vec{B})$

The potential difference between the ends P and Q is

$$V = \vec{E}.\vec{l} = (\vec{v}\times\vec{B}).\vec{l}$$

It is the magnetic force on the moving free electrons that maintains the potential difference and produces the emf $E = B\ell v$.

For $\vec{B}\perp\vec{v}\perp\vec{\ell}$

As this emf is produced due to the motion of a conductor, it is called a motional emf. The concept of motional emf for a conductor can be generalized for any shape moving in any magnetic field uniform or not. For an element $d\vec{l}$ of conductor the contribution due to the emf of the magnitude

$$de = (\vec{v}\times\vec{B}).d\vec{l}$$

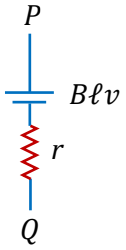
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Therefore total EMF across any arbitrary shape wire is given by

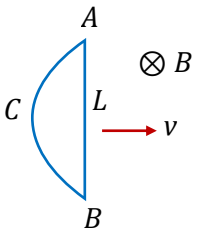
$$E = \int(\vec{v}\times\vec{B}).d\vec{l}$$

Motional emf across an arbitrary shape wire moving translationally in uniform magnetic field:

From previous section it can be said that a moving rod is electrically equivalent to the following diagram.



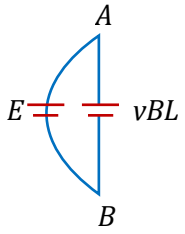
Now consider a situation as shown by the Figure in which a closed coil $ABCA$ is moving in a uniform magnetic field $\boldsymbol{B}$ with velocity $\boldsymbol{v}$. The flux passing through the coil is constant and therefore the induced emf is zero.



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Now consider rod AB, which is a part of the coil. Emf induced in the rod = $B\,L\,v$

Suppose the emf induced in part ACB is E, as shown.



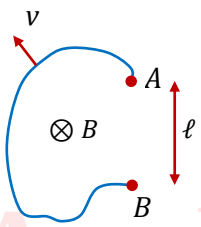
Since the emf in the coil is zero, Emf (in ACB) + Emf (in BA) = 0

or $\quad -E + vBL = 0 \quad$ or $\quad E = vBL$

Thus, emf induced in any path joining A and B is same, provided the magnetic field is uniform. Also, the equivalent emf between A and B is BLv (here the two emf's are in parallel)

Illustration 4

Figure shows an irregular shaped wire AB moving with velocity v, as shown.



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Find the emf induced in the wire.

Solution:

The same emf will be induced in the straight imaginary wire joining A and B, which is $Bv\ell \sin\theta$

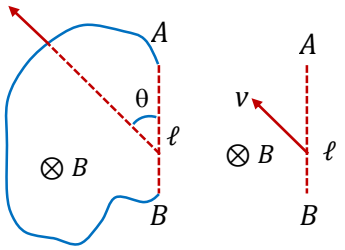
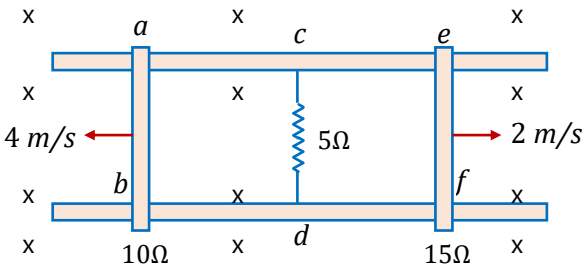


Illustration 5

Two parallel rails with negligible resistances are 10 *cm* apart. They are connected by a 5 Ω resistor. The circuit also contains two metal rods having resistance of 10 Ω and 15 Ω along the rails (fig). The rods are pulled away from the resistor at constant speeds 4 *m/s* and 2 *m/s* respectively. A uniform magnetic field of magnitude 0.01 *T* is applied perpendicular to the plane of the rails. Determine the current in the 5 Ω resistor.



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Solution:

As two conductors are moving in uniform magnetic field, motional emf will be induced across them. The rod ab will act as a source of emf $e_1 = Bv\ell = (0.01)(4)(0.1) = 4\times10^{-3}V$

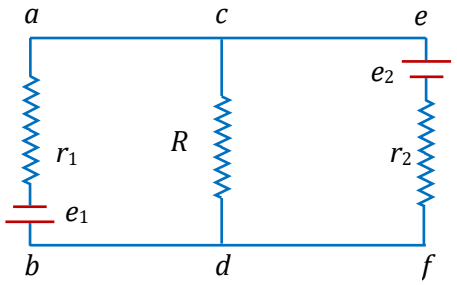
And it's internal resistance r_1 = 10 Ω

Similarly, rod *ef* will also act as source of *emf*,

$$e_2 = (0.01)(2)(0.1) = 2\times10^{-3}V$$

with internal resistance r_2 = 15 Ω

From right hand rule: $V_b > V_a$ and $V_e > V_f$



Also R = 5 Ω,

$$E_{eq} = \frac{e_1 r_2 - e_2 r_1}{r_1 + r_2} = \frac{60\times10^{-3} - 20\times10^{-3}}{15+10} = \frac{40}{25}\times10^{-3} = 1.6\times10^{-3}\,volt$$

$$r_{eq} = \frac{15\times10}{15+10} = 6\Omega$$

and $$I = \frac{E_{eq}}{r_{eq}+R} = \frac{1.6\times10^{-3}}{6+5} = \frac{1.6}{11}\times10^{-3} = \frac{8}{55}\times10^{-3}\,Amp$$ from *d* to *c*

Illustration 6

A rod of length *l* is kept parallel to a long wire carrying constant current *i*. It is moving away from the wire with a velocity *v*. Find the emf induced in the wire when its distance from the long wire is *x*.

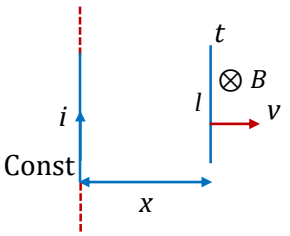
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Solution:

$$E = Blv = \frac{\mu_0 ilv}{2\pi x}$$

OR

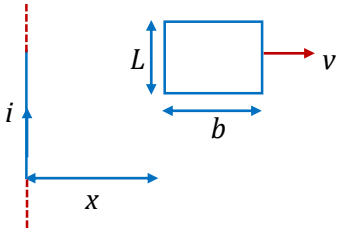
Emf is equal to the rate with which magnetic field lines are cut. In *dt* time the area swept by the rod is *l v dt*. The magnetic field lines cut in *dt* time = $Blvdt = \frac{\mu_0 ilvdt}{2\pi x}$.



$\therefore$ The rate with which magnetic field lines are cut = $\frac{\mu_0 ilv}{2\pi x}$

Illustration 7

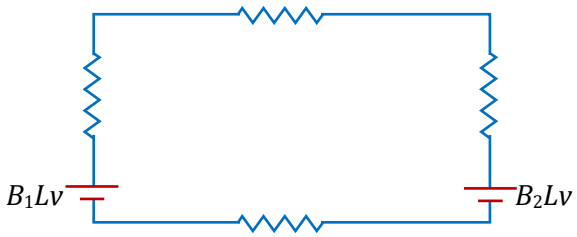
A rectangular loop, as shown in the figure, moves away from an infinitely long wire carrying a current *i*. Find the emf induced in the rectangular loop.



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Solution:

$$E = B_1Lv - B_2Lv = \frac{\mu_0 i}{2\pi x}Lv - \frac{\mu_0 i}{2\pi(x+b)}Lv = \frac{\mu_0 iLbv}{2\pi x(x+b)}$$



Alter:

Consider a small segment of width dy at a distance y from the wire.

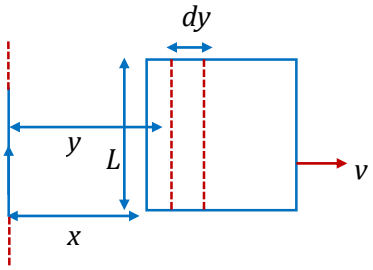
Let flux through the segment be $d\phi$.

$$\therefore \quad d\phi = \frac{\mu_0 i}{2\pi y}Ldy$$

$$\therefore \quad \phi = \frac{\mu_0 iL}{2\pi}\int_x^{x+b}\frac{dy}{y} = \frac{\mu_0 iL}{2\pi}\left(\ell n(x+b) - \ell nx\right)$$

Now $$\frac{d\phi}{dt} = \frac{\mu_0 iL}{2\pi}\left[\frac{1}{x+b}\frac{dx}{dt} - \frac{1}{x}\frac{dx}{dt}\right]v = \frac{-\mu_0 ibLv}{2\pi x(x+b)}$$

Constant current i

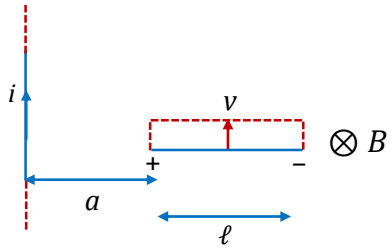


$$\therefore \quad \text{Induced emf} = \frac{\mu_0 ibLv}{2\pi x(x+b)}$$

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Illustration 8

A rod of length l is placed perpendicular to a long wire carrying current i. The rod is moved parallel to the wire with a velocity v. Find the emf induced in the rod, if its nearest end is at distance *'a'* from the wire.



Solution:

Consider a segment of the rod of length dx, at a distance x from the wire. Emf induced in the segment

$$dE = \frac{\mu_0 i}{2\pi x}dx.v$$

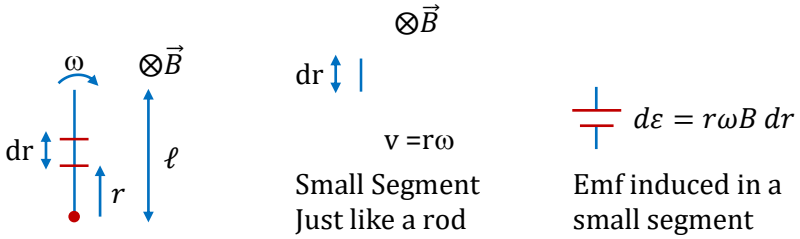
$$\therefore \quad E = \int_a^{a+\ell}\frac{\mu_0 ivdx}{2\pi x} = \frac{\mu_0 iv}{2\pi}\ell n\left(\frac{\ell+a}{a}\right)$$

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Induced emf due to rotation

Rotation of the rod

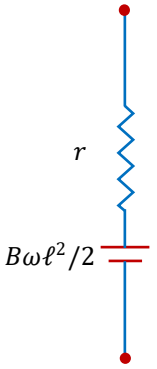
Consider a conducting rod of length l rotating in a uniform magnetic field.



Emf induced in a small segment of length dr of the rod = $v\,B\,dr = r\omega\,B\,dr$

$\therefore$ emf induced in the rod = $\omega B\int_0^{\ell} r dr = \frac{1}{2}B\omega\ell^2$

Equivalent of this rod is as following (figure):



or $\varepsilon = \frac{d\Phi}{dt}$

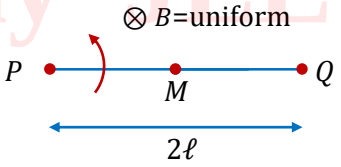
$\varepsilon = \frac{d\Phi}{dt} = \frac{\text{Flux through the area swept by the rod in time dt}}{dt}$

$= \frac{B\frac{1}{2}\ell^2\omega dt}{dt} = \frac{1}{2}B\omega\ell^2$

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Illustration 9

A rod PQ of length 2ℓ is rotating about one end P in a uniform magnetic field B which is perpendicular to the plane of rotation of the rod. Point M is the mid point of the rod. Find the induced emf between M and Q if that between P and Q = 100V.



Solution:

$E_{MQ} + E_{PM} = E_{PQ}$

$E_{PM} = \frac{Bw\ell^2}{2}$

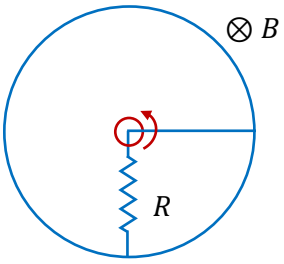
$E_{PQ} = \frac{1}{2}B\omega(2\ell)^2 = 100$

$B\omega\ell^2 = 50$

$E_{MQ} = E_{PQ} - E_{PM} = 100 - \frac{50}{2} = 75V$

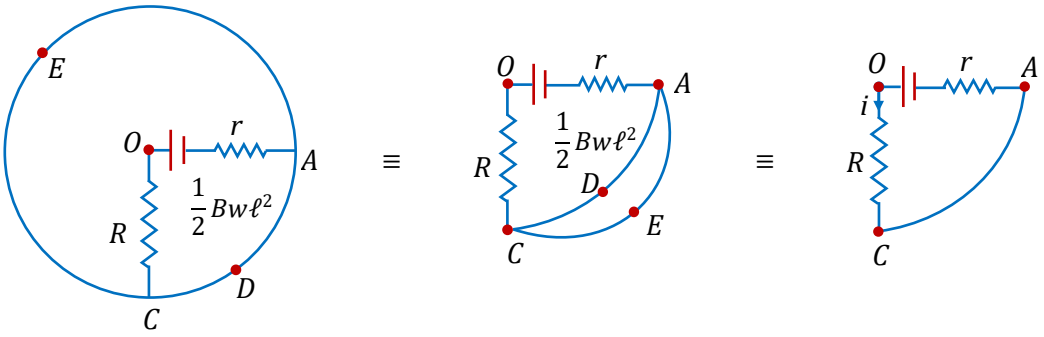
Illustration 10

A rod of length ℓ and resistance r rotates about one end as shown in figure. Its other end touches a conducting ring of negligible resistance. A resistance R is connected between centre and periphery. Draw the electrical equivalence and find the current in the resistance R. There is a uniform magnetic field B directed as shown.



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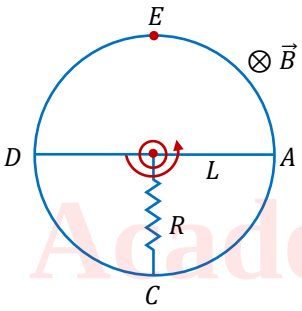
Solution:



current $i=\dfrac{\frac{1}{2}B\omega\ell^2}{R+r}$

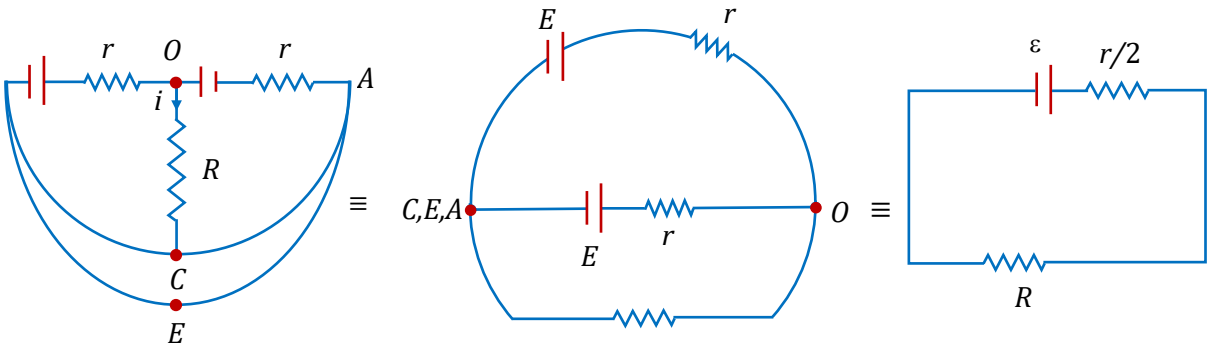
Illustration 11

Solve the above question if the length of rod is $2L$ and resistance $2r$ and it is rotating about its centre. Both ends of the rod now touch the conducting ring.



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Solution:

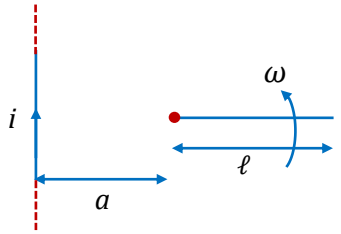


$$i=\frac{\varepsilon}{R+\frac{r}{2}}=\frac{\frac{1}{2}B\omega L^2}{R+\frac{r}{2}}$$

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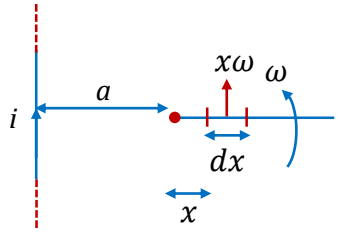
Illustration 12

A rod of length l is rotating with an angular speed ω about its one end which is at a distance *'a'* from an infinitely long wire carrying current i. Find the emf induced in the rod at the instant shown in the figure.



Solution:

Consider a small segment of rod of length dx, at a distance x from one end of the rod. Emf induced in the segment



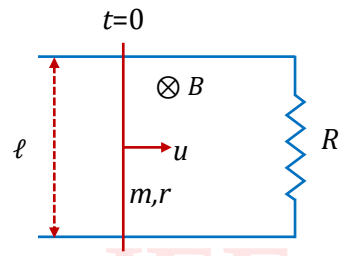
$$dE = \frac{\mu_0 i}{2\pi(x+a)}(x\omega)dx$$

$$\therefore \quad E = \int_0^{\ell} \frac{\mu_0 i}{2\pi(x+a)}(x\omega)dx = \frac{\mu_0 i\omega}{2\pi}\left[\ell - a.\ell n\left(\frac{\ell+a}{a}\right)\right]$$

Velocity, force and power analysis for a conductor moving in a magnetic field:

Illustration 13

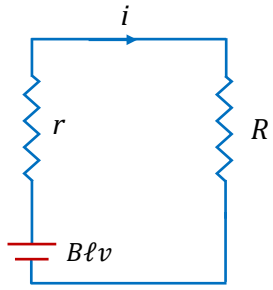
A rod of mass m and resistance r is placed on fixed, resistance less, smooth conducting rails (closed by a resistance R) and it is projected with an initial velocity u. Find its velocity as a function of time.



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Solution:

Let at an instant the velocity of the rod be v. The emf induced in the rod will be $B\ell v$. The electrically equivalent circuit is shown in the following diagram.



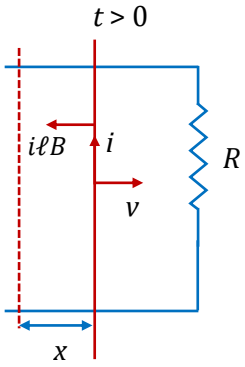
$\therefore$ Current in the circuit,

$$i = \frac{B\ell v}{R+r}$$

At time t magnetic force acting on the rod is $F = i\ell B$, opposite to the motion of the rod.

$$i\ell B = -m\frac{dv}{dt} \qquad ...(1)$$

$$i = \frac{B\ell v}{R+r} \qquad ...(2)$$



Now solving these two equations

$$\frac{B^2\ell^2 v}{R+r} = -m.\frac{dv}{dt}$$

$$-\frac{B^2\ell^2}{(R+r)m}.dt = \frac{dv}{v}$$

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Let $\frac{B^2\ell^2}{(R+r)m}=k$

$-K.dt=\frac{dv}{v}$

$\int_u^v \frac{dv}{v}=\int_0^t -K.dt$

$\ln\left(\frac{v}{u}\right)=-Kt$

$v = ue^{-Kt}$

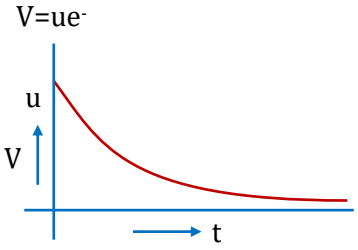


Illustration 14

In the above question find the force required to move the rod with constant velocity v, and also find the power delivered by the external agent.

Solution:

The force needed to keep the velocity constant,

$$F_{ext}=i\ell B=\frac{B^2\ell^2 v}{R+r}$$

Power due to external force = $\frac{B^2\ell^2v^2}{R+r}=\frac{\varepsilon^2}{R+r}=i^2(R+r)$

Note:

The power delivered by the external agent is converted into joule heating in the circuit. That means magnetic field helps in converting the mechanical energy into joule heating.

Illustration 15

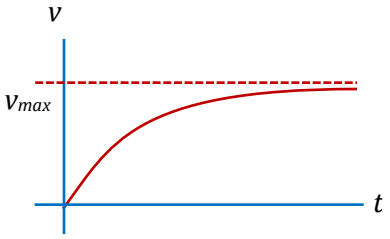
In the above question if a constant force F is applied on the rod. Find the velocity of the rod as a function of time, assuming it started with zero initial velocity.

Solution:

$m\frac{dv}{dt}=F-i\ell B$ ...(1)

$i=\frac{B\ell v}{R+r}$ ...(2)

$m\frac{dv}{dt}=F-\frac{B^2\ell^2 v}{R+r}$



Let $K=\frac{B^2\ell^2}{R+r}$ $\Rightarrow$ $\int_0^v \frac{dv}{F-Kv}=\int_0^t \frac{dt}{m}$

$-\frac{[\ell n(F-Kv)]_0^v}{K}=\frac{t}{m}$ $\Rightarrow$ $\ell n\left(\frac{F-Kv}{F}\right)=-\frac{Kt}{m}$

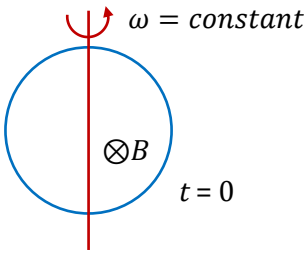
$F-Kv=Fe^{-kt/m}$ $\Rightarrow$ $v=\frac{F}{K}\left(1-e^{-kt/m}\right)$

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EMF Induced due to rotation of a coil

Illustration 16

A ring rotates with angular velocity ω about an axis in the plane of the ring and which passes through the center of the ring. A constant magnetic field B exists perpendicular to the plane of the ring. Find the emf induced in the ring as a function of time.



Solution:

At any time, t, $\phi = BA\cos\theta = BA\cos\omega t$

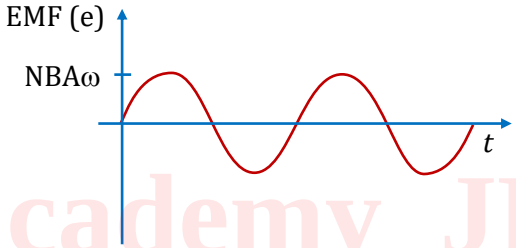
Now induced emf in the loop

$$e = \frac{-d\phi}{dt} = BA\omega\sin\omega t$$

If there are N turns

$$\text{emf} = BA\omega N\sin\omega t$$

$BA\omega N$ is the amplitude of the emf



$$e = e_m\sin\omega t$$

$$i = \frac{e}{R} = \frac{e_m}{R}\sin\omega t = i_m\sin\omega t$$

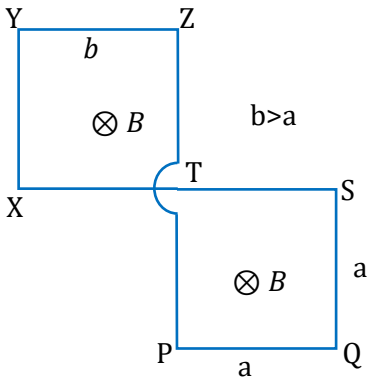
$$i_m = \frac{e_m}{R}$$

The rotating coil thus produces a sinusoidally varying current or alternating current. This is also the principle used in generator.

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Illustration 17

Figure shows a wire frame $PQSTXYZ$ placed in a time varying magnetic field given as $B = \beta t$, where β is a positive constant. Resistance per unit length of the wire is λ. Find the current induced in the wire and draw its electrical equivalent diagram.



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Solution:

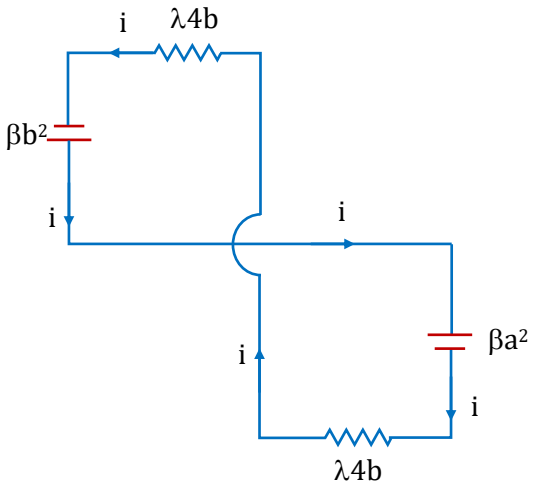
Induced emf in part $PQST = \beta a^2$ (in anticlockwise direction, from Lenz's Law)

Similarly Induced emf in part $TXYZ = \beta b^2$

(in anticlockwise direction, from Lenz's Law)

Total resistance of the part $PQST = \lambda 4a$.

Total resistance of the part $TXYZ = \lambda 4b$. The equivalent circuit is as shown in the following diagram.



Writing *KVL* along the current flow

$$\beta b^2 - \beta a^2 - \lambda\, 4ai - \lambda\, 4bi = 0$$

$$i = \frac{\beta}{4\lambda}(b-a)$$

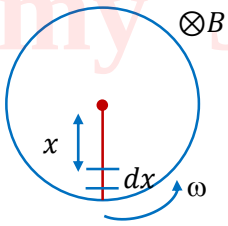
Emf induced in a rotating disc:

Consider a disc of radius r rotating in a magnetic field B.

Consider an element dx at a distance x form the centre. This element is moving with speed $v = \omega x$.

$\therefore$ Induced emf across dx

$= B(dx)\, v = Bdx\omega x = B\omega x dx$



$\therefore$ Emf between the centre and the edge of disc.

$$= \int_0^r B\omega x dx = \frac{B\omega r^2}{2}$$

Induced Electric Field

If $\vec{E}$ be the induced electric field, the force on a charge q placed in field is $q\vec{E}$. The work done per unit charge as the charge moves through $d\vec{l}$ is $\vec{E}.d\vec{l}$.

The emf developed in the loop is, therefore,

$$\varepsilon = \oint \vec{E}.d\vec{l}$$

Using Faraday's law of induction,

$$\varepsilon = -\frac{d\phi}{dt} \qquad \text{or} \qquad \oint \vec{E}.d\vec{l} = -\frac{d\phi}{dt}$$

The presence of a conducting loop is not necessary to have an induced electric field. As long as $\vec{B}$ keeps changing, the induced electric field is present. If a loop is there, the free electrons start drifting and consequently an induced current results.

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Calculating the Induced Electric Field

The induced electric field is peculiar in another way: It is nonconservative. Recall that a force is conservative if it does no net work on a particle moving around a closed path. "Uphill" are balanced by "downhills." We can associate a potential energy with a conservative force, hence we have gravitational potential energy for the conservative gravitational force and electric potential energy for the conservative electric force of charges (Coulomb electric field).

But a charge moving around a closed path in the induced electric field is always being pushed in the same direction by the electric force $F = qE$. There's never any negative work to balance the positive work, so the net work done in going around a closed path is not zero. Because it is nonconservative, we cannot associate an electric potential with an induced electric field. Only the Coulomb field of charges has an electric potential.

However, we can associate the induced field with the emf of Faraday's law. The emf was defined as the work required per unit charge to separate the charge. That is,

$$\varepsilon = \frac{W}{q}$$

In batteries this work, a familiar source of emf, is done by chemical forces. But the emf that appears in Faraday's law arises when work is done by the force of an induced electric field.

If a charge q moves through a small displacement $d\vec{s}$, the small amount of work done by the electric field is $dW = \vec{F}.d\vec{s} = q\vec{E}.d\vec{s}$. The emf of Faraday's law is an emf around a closed curve through which the magnetic flux ϕ_m is changing. The work done by the induced electric field as charge q moves around a closed curve is

$$W_{\text{closed curve}} = q\oint \vec{E}.d\vec{s}$$

Where the integration symbol with the circle is the same as the one we used in Ampere's law to indicate an integral around a closed curve. If we use this work in equation, we find that the emf around a closed loop is

$$\varepsilon = \frac{W_{\text{closed curve}}}{q} = \oint \vec{E}.d\vec{s}$$

If we restrict ourselves to situations such as figure where the loop is perpendicular to the magnetic field and only the field is changing, we can write Faraday's law as $\varepsilon = |d\phi_m / dt| = A|dB/dt|$. Consequently

$$\oint \vec{E}.d\vec{s} = A\left|\frac{dB}{dt}\right|$$

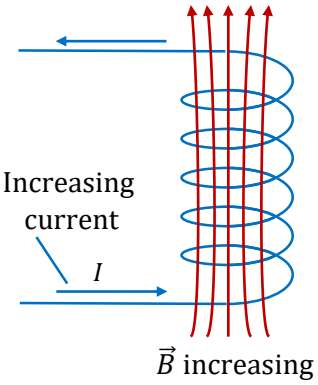
Above equation is an alternative of statement of Faraday's law that relates the induced electric field to the changing magnetic field.

The solenoid in figure (a) provides a good example of the connection between $\vec{E}$ and $\vec{B}$. If there were a conduction loop inside the solenoid, we could use Lenz's law to determine that the direction of the induced current would be clockwise. But Faraday's law, in the form of above equation, tells us that an induced electric field is present whether there is a conducting loop or not. The electric field is induced simply due to the fact that $\vec{B}$ is changing.

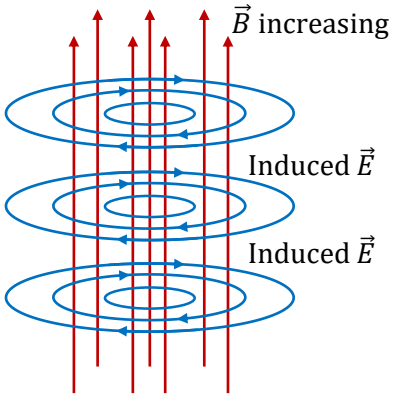
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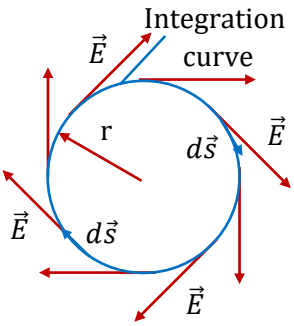
(a) The current through the solenoid is increasing.



(b) The induced electric field circulates around the magnetic field lines.



(c) Top view into the solenoid $\vec{B}$ is coming out of the page.



In figure, the induced electric field circulates around the changing magnetic field inside a solenoid.

The shape and direction of the induced electric field have to be such that it could drive a current around a conducting loop, if one were present and it has to be consistent with the cylindrical symmetry of the solenoid. The only possible choice, shown in Figure is an electric field that circulates clockwise around the magnetic field lines.

Note:

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Circular electric field lines violate the rule that electric field lines have to start and stop on charges. However, that rule applied only to Coulomb fields created by source charges. An induced electric field is a non-Coulomb field created not by source charges but by a changing magnetic field. Without source charges, induced electric field lines must form closed loops.

To use Faraday's law, choose a clockwise circle of radius r as the closed curve for evaluating the integral. Figure (c) shows that the electric field vectors are everywhere along the tangent to the curve, so the line integral of $\vec{E}$ is

$$\oint \vec{E}.d\vec{s} = El = 2\pi rE$$

Where $l = 2\pi r$ is the length of the closed curve. This is exactly like the integrals we did for Amperes law.

If we stay inside the solenoid ($r < R$), the flux passes through area $A = \pi r^2$ and above equation becomes

$$\oint \vec{E}.d\vec{s} = 2\pi rE = A\left|\frac{dB}{dt}\right| = \pi r^2\left|\frac{dB}{dt}\right|$$

Illustration 18

A 4 *cm* diameter solenoid is wound with 2000 turns per meter. The current through the solenoid oscillates at 60 *Hz* with an amplitude of 2 *A*. What is the maximum strength of the induced electric field inside the solenoid?

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Solution:

You learned in that the magnetic field strength inside a solenoid with n turns per meter is $B = \mu_0 nI$. In this case, the current through the solenoid is $I = I_0 \sin \omega t$, where $I_0 = 2\ A$ is the peak current and $\omega = 2\pi(60Hz) = 377\ rad/s$. Thus, the induced electric field strength at radius r is

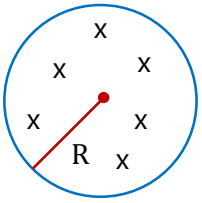
$$E = \frac{r}{2}\left|\frac{dB}{dt}\right| = \frac{r}{2}\frac{d}{dt}\left(\mu_0 nI_0 \sin\omega t\right) = \frac{1}{2}\mu_0 nr\omega I_0 \cos\omega t$$

The field strength is maximum at maximum radius ($r = R$) and at the instant when $\cos \omega t = 1$. That is,

$$E_{max} = \frac{1}{2}\mu_0 nR\omega I_0 = 0.019 V/m$$

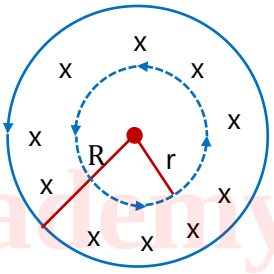
Illustration 19

The magnetic field at all points within the cylindrical region whose cross-section is indicated in the figure start increasing at a constant rate $\alpha \frac{\text{tesla}}{\text{second}}$. Find the magnitude of electric field as a function of r, the distance from the geometric centre of the region.



Solution:

For $r \le R$:



$$\because \quad E\ell = A\left|\frac{dB}{dt}\right|$$

$$\therefore \quad E(2\pi r) = (\pi r^2)\alpha$$

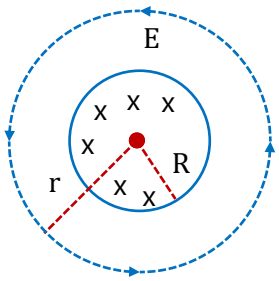
$$\Rightarrow \quad E = \frac{r\alpha}{2} \quad \Rightarrow \quad E \propto r$$

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E-r graph is straight line passing through origin.

At $r = R,\ E = \frac{R\alpha}{2}$

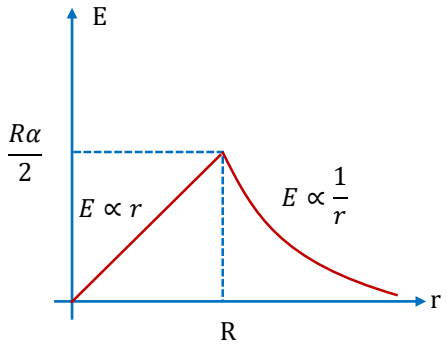
For $r \ge R$



$$\because \quad E\ell = A\left|\frac{dB}{dt}\right|$$

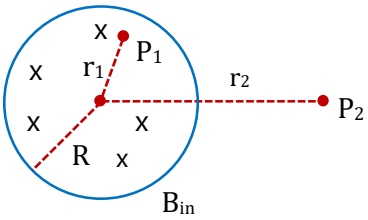
$$\therefore \quad E(2\pi r) = (\pi R^2)\alpha$$

$$\Rightarrow \quad E = \frac{\alpha R^2}{2r} \Rightarrow E \propto \frac{1}{r}$$



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Illustration 20



For the situation described in figure the magnetic field changes with time according to, $B=\left(2t^3-4t^2+0.8\right)T$ and $r_2=2R=5cm$

(a) Calculate the force on an electron located at P_2 at $t = 2s$

(b) What are the magnitude and direction of the electric field at P_1 when $t = 3$ s and $r_1 = 0.02m$?

Solution:

$$E\ell=A\left|\frac{dB}{dt}\right|\Rightarrow E=\frac{\pi R^2}{2\pi r_2}\frac{d}{dt}\left(2t^3-4t^2+0.8\right)=\frac{R^2}{2r_2}\left(6t^2-8t\right)$$

(a) Force on electron at P_2 is $F = eE$

At $t=2s,\ F=\dfrac{1.6\times10^{-19}\times\left(2.5\times10^{-2}\right)^2}{2\times5\times10^{-2}}\times\left[6(2)^2-8(2)\right]$

$$=\frac{1.6}{4}\times2.5\times10^{-21}\times(24-16)=8\times10^{-21}N \text{ at } t=2s$$

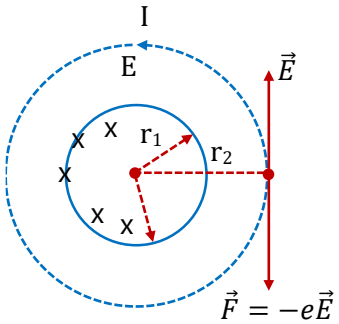
$\dfrac{dB}{dt}$ is positive so it is increasing.

$\therefore$ Direction of induced current and E are as shown in figure and hence force on electron having charge-e is right perpendicular to r_2 downwards.

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(b) For $r_1 = 0.02m$ and at $t = 3s$,

$$E=\frac{\pi r_1^2}{2\pi r_1}\left(6t^2-8t\right)=\frac{0.02}{2}\times\left[6(3)^2-8(3)\right]=0.3V/m$$

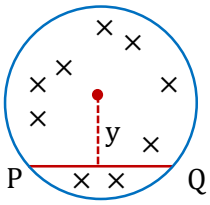


At $t=3\text{sec}$, $\dfrac{dB}{dt}$ is positive, so B is increasing and hence direction of E is same as in case (a) and it is left perpendicular to r_1 upward.

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Illustration 21

For the situation described in figure the magnetic field changes with time according to $B = kt$. Find the value of induced electric field across PQ.



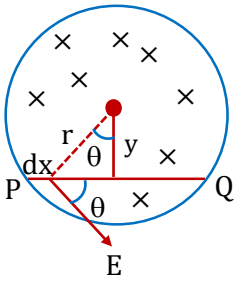
Solution:

M-1: $dE = \vec{E}.d\vec{x}$

$$dE = \frac{k}{2}(r.\cos\theta).dx$$

$$\Rightarrow \quad \int dE = \int \frac{ky}{2}dx = \frac{ky}{2} \times \ell$$

$$E_{PQ} = \frac{ky\ell}{2}$$

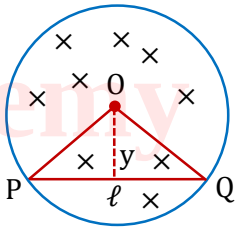


M-2: $\phi_{OPQ} = B \times \frac{1}{2}(y\ell)$

$$E_{OPQ} = \frac{d\phi}{dt} = \frac{ky\ell}{2}$$

$$E_{OP} = +E_{PQ} + E_{QO} = \frac{ky\ell}{2}$$

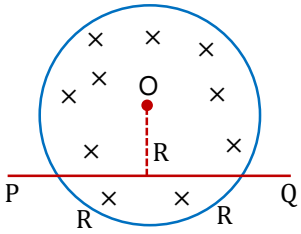
$$E_{PQ} = \frac{ky\ell}{2} \qquad (\because E_{OP} = E_{QO} = 0)$$



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Illustration 22

For the situation described in figure the magnetic field changes with time according to $B = kt$. Find the value of induced electric field across PQ.



Solution:

$$E = \frac{d\phi}{dt} \quad \Rightarrow \quad E = \frac{dB}{dt} \times A$$

$$E_{OP} + E_{PQ} + E_{QO} = k \times \frac{\pi R^2}{4}$$

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Self-Inductance:

In the absence of magnetic materials, the *B*-field at any point due to a single circuit of given shape is proportional to the current in it, so the current is measured by it magnetic effect. It follows that the total flux ϕ linking a circuit due to its own current *I* is proportional to *I*. We define the self inductance *L* of a circuit as the ratio of ϕ to *I* so that

$$\boxed{\phi = LI \quad \text{(definition of L)}}$$

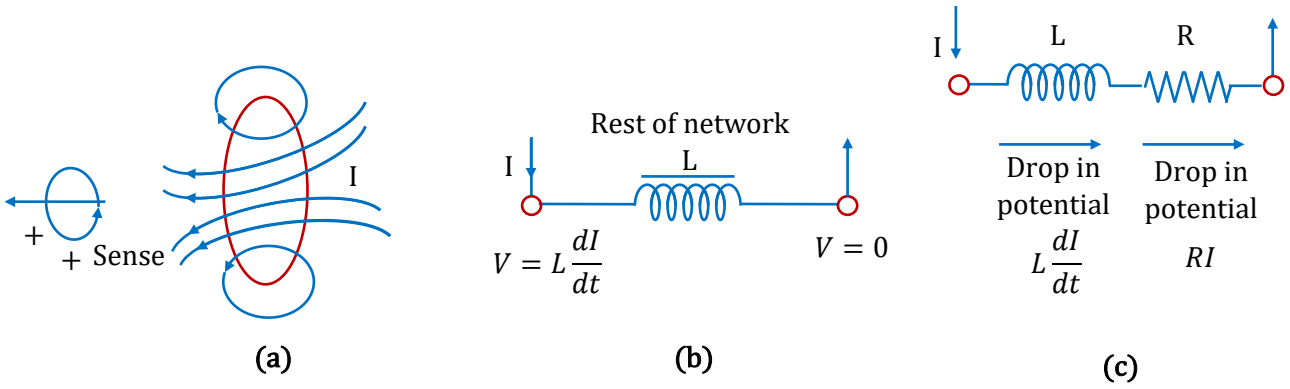


Figure: (a) Self-flux of a circuit: increase of flux due to an increase in I causes an electromotance in the negative sense and this opposes the growth of I; (b) self-inductance as a circuit element; (c) the equivalent circuit of a self-inductor with resistance.

And *L* is thus a constant for a circuit of given shape and size. The self-flux of a circuit defines the positive direction for ϕ so that *L* is essentially positive. Its SI unit, the WbA^{-1}, is called the henry, symbol *H*.

Self-induction when the current in a rigid circuit changes, the induced electromotance is

$$\boxed{\varepsilon = -LdI/dt}$$

The negative sign indicating that when *I* increases, ε is negative and is thus in a direction opposing the growth of *I* (fig. a). When a device is specially constructed so that above equation applies, it is known as an inductor and has the symbol as in Fig. b in a circuit diagram, crossed by an arrow if variable and with parallel lines adjacent to it if it has a ferromagnetic core.

In a network with varying currents, an inductor acts as an electromotance – *LdI/dt* with an internal resistance which cannot be avoided. Like a cell, therefore, it can be represented by an equivalent network consisting of a pure inductance *L* in series with its resistance *R* as in Fig. c. The pure inductance has no resistance and thus has a potential difference across its ends equal to the electromotance: once the direction of I has been chosen arbitrarily in any problem the potential drop across an inductor *L* is *LdI/dt* just as that across *R* is *RI* (fig. c).

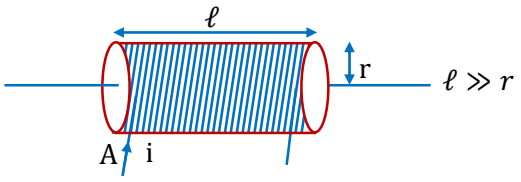
Self inductance of solenoid:

Let the volume of the solenoid be *V*, the number of turns per unit length be *n*.

Let a current *i* be flowing in the solenoid.

Magnetic field in the solenoid is given as $B = \mu_0 ni$.

The magnetic flux through one turn of solenoid $\phi = \mu_0 niA$.



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The total magnetic flux through the solenoid = $N\phi = N\mu_0 ni\, A = \mu_0 n^2 i\, A\, \ell$

$\therefore \quad L = \mu_0 n^2 \ell A = \mu_0 n^2 V \quad \Rightarrow \quad \phi = \mu_0\, n\, i\, \pi r^2\, (n\ell) \quad \Rightarrow \quad L = \frac{\phi}{i} = \mu_0\, n^2\, \pi r^2 \ell$

Inductance per unit volume = $\mu_0 n^2$.

Self inductance is the physical property of the loop due to which it opposes the change in current that means it tries to keep the current constant. Current cannot change suddenly in the inductor.

Self-Inductance of Toroid:

$$L = \frac{N\phi}{I} = \frac{N}{I}\int \vec{B}.d\vec{A}$$

$$\int \vec{B}.d\vec{A} = \int B.dA = \int_a^b \frac{\mu_0 NI}{2\pi x} h dx = \frac{\mu_0 NIh}{2\pi} \ell n\left(\frac{b}{a}\right)$$

$$L = \frac{\mu_0 N^2}{2\pi} h\, \ell n\left(\frac{b}{a}\right)$$

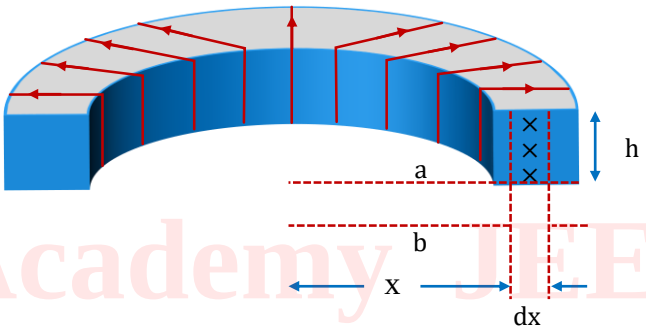
Let, $b - a = d$

$$\ell n\left(\frac{b}{a}\right) = \ell n\left(1 + \frac{b-a}{a}\right)$$

If, $b - a << a$

$b - a << b$

Then, $\ell n\left(1 + \frac{b-a}{a}\right) \simeq \left(\frac{b-a}{a}\right)$



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$$L = \frac{\mu_0 N^2}{2\pi} \frac{h(b-a)}{a} = \frac{\mu_0 N^2 A}{\ell} = \mu_0 n^2 \ell A$$

$$\frac{L}{\ell} = \mu_0 n^2 A$$

which is Inductance per unit length of infinitely long solenoid.

Note:

If there is a resistance in the inductor (resistance of the coil of inductor) then :

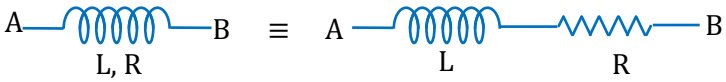
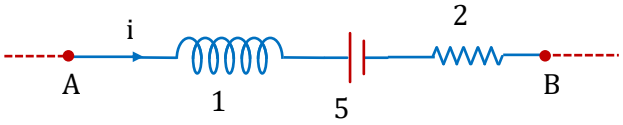


Illustration 23

A B is a part of circuit. Find the potential difference $V_A - V_B$ if



(i) current $i = 2A$ and is constant.

(ii) current $i = 2A$ and is increasing at the rate of 1 amp/sec.

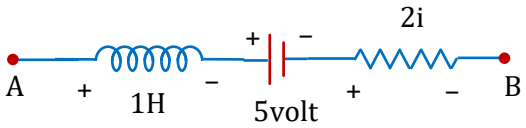
(iii) current $i = 2A$ and is decreasing at the rate 1 amp/sec.

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Solution:

$$L\frac{di}{dt}=1\frac{di}{dt}$$

Writing *KVL* from *A* to *B*



$$V_A-1\frac{di}{dt}-5-2i=V_B$$

(i) Put $i=2,\frac{di}{dt}=0$

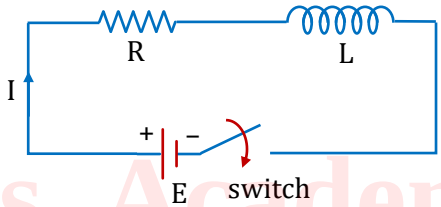
$V_A-5-4=V_B$ $\quad\therefore\quad V_A-V_B=9volt$

(ii) Put $i=2,\frac{di}{dt}=1;V_A-1-5-4=V_B$ or $V_A-V_B=10volt$

(iii) Put $i=2,\frac{di}{dt}=-1;V_A+1-5-2\times2=V_B$ or $V_A-V_B=8Volt$

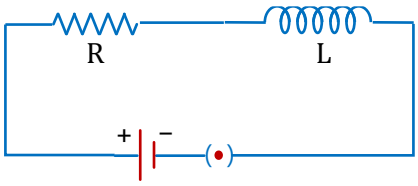
R-L DC circuit:

Current Growth



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(i) Emf equation $E=IR+L\frac{dl}{dt}$

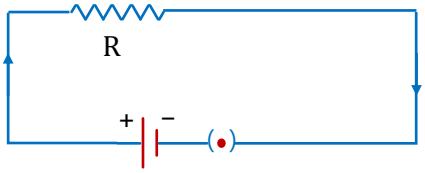


(ii) Current at any instant

When key is closed the current in circuit increases exponentially with respect to time. The current in circuit at any instant '*t*' is given by $I=I_0\left[1-e^{\frac{-t}{\lambda}}\right]$

$T=0$ (just after the closing of key) $\Rightarrow I=0$

$T=\infty$ (sometime after closing of key) $\Rightarrow I\to I_0$



(iii) Just after the closing of the key inductance behaves like open circuit and current in circuit is zero. Open circuit, $t=0, I=0$, inductor provide infinite resistance.

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(iv) Sometime after closing of the key inductance behaves like simple connecting wire (short circuit) and current in circuit is constant.

Short circuit, $t \to \infty, I \to I_0$. Inductor provide zero resistance $I_0 = \frac{E}{R}$

(Final, steady, maximum or peak value of current) or ultimate current.

Note:

Peak value of current in circuit does not depends on self inductance of coil.

(v) Time constant of circuit (λ)

$$\lambda = \frac{L}{R_{sec}}$$

It is a time in which current increases up to 63% or 0.63 times of peak current value.

(vi) Rate of growth of current at any instant:

$$\left[\frac{dI}{dt}\right] = \frac{E}{L}\left(e^{-t/\lambda}\right)$$

$$\Rightarrow \quad t = 0 \quad \Rightarrow \quad \left[\frac{dI}{dt}\right]_{max} = \frac{E}{L}$$

$$t = \infty \quad \Rightarrow \quad \left[\frac{dI}{dt}\right] \to 0$$

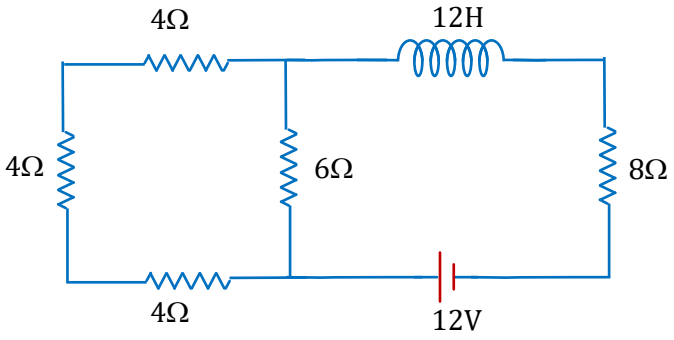
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Note:

Maximum or initial value of rate of growth of current does not depends upon resistance of coil.

Illustration 24

For the circuit shown, which of the following statement(s) is (are) correct?



(A) Its time constants is 2 second.

(B) In steady state, current through inductance will be 1A.

(C) In steady state, current through 4Ω resistance will be 2/3A.

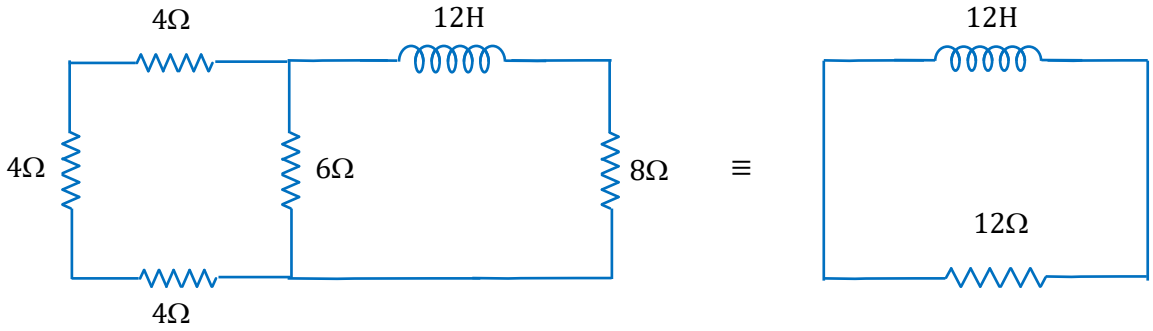
(D) In steady state, current through 8Ω resistance will be zero.

Ans. (B)

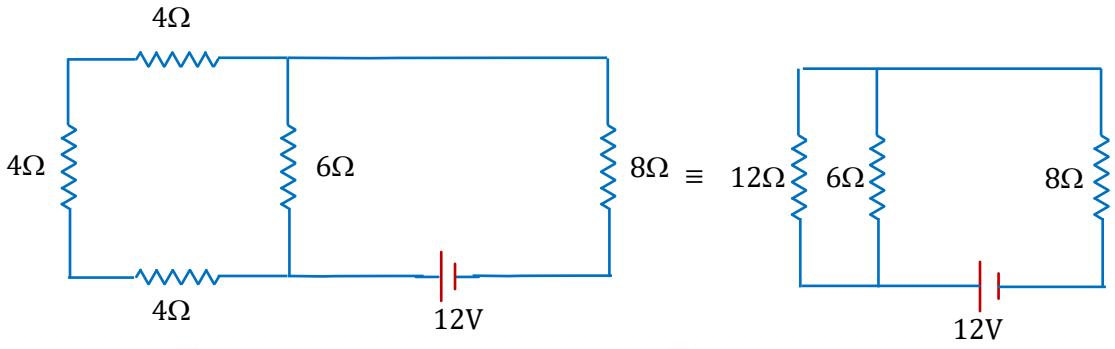
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Solution:

Time constant $\tau = \frac{L}{R} = \frac{12}{12} = 1s$



In steady state



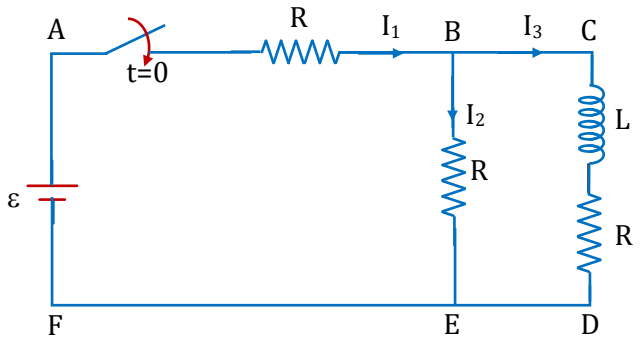
Current through 4Ω resistance = $\frac{6}{6+12} \times 1 = \frac{1}{3} A$

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Illustration 25

In the following circuit, the switch is close at $t = 0$. Find the currents i_1, i_2, i_3 and $\frac{di_3}{dt}$ at $t = 0$ and at $t = \infty$.

Initially all current are zero.



Solution:

At $t = 0$

i_3 is zero, since current cannot suddenly change due to the inductor.

$\therefore \quad i_1 = i_2$ (From KCL)

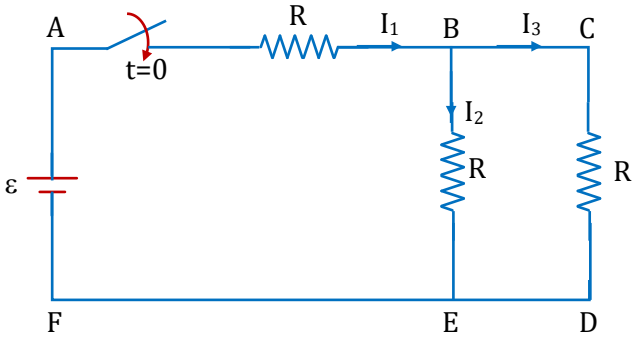
Applying KVL in the part ABEF we get, $i_1 = i_2 = \frac{\varepsilon}{2R}$

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Also $\frac{di}{dt}=\frac{\varepsilon}{L}$

At $t=\infty$

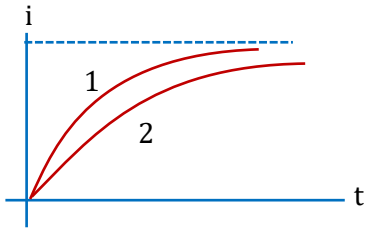
i_3 will become constant and hence potential difference across the inductor will be zero. It is just like a simple wire and the circuit can be solved assuming it to be like shown in the following diagram.



$$i_2=i_3=\frac{\varepsilon}{3R}, i_1=\frac{2\varepsilon}{3R}, \text{ also } \frac{di_3}{dt}=0$$

Illustration 26

Which of the two curves shown has less time constant.



Ans. Curve 1

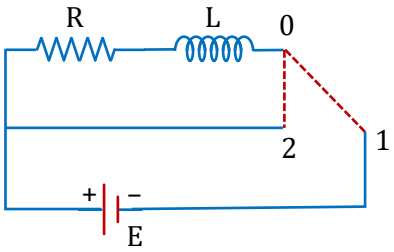
Solution:

$$i=i_0\left(1-e^{-t/\lambda}\right)\Rightarrow \frac{di}{dt}=\frac{i_0}{\lambda}=slope$$

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Current Decay:

(i) **Emf equation** $IR+L\frac{di}{dt}=0$



(ii) **Current at any instant**

Once current acquires its final max steady value, if suddenly switch is put off then current start decreasing exponentially wrt to time. At switch put off condition $t=0, I=I_0$, source emf E is cut off from circuit $I=I_0\left(e^{-t/\lambda}\right)$

Just after opening of key $t=0$ $\Rightarrow$ $I=I_0=\frac{E}{R}$

Some time after opening of key $t\to\infty$ $\Rightarrow$ $I\to 0$

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(iii) Time constant (λ)

It is a time in which current decreases up to 37% or 0.37 times of peak current value.

(iv) Rate of decay of current at any instant

$$\left[-\frac{di}{dt}\right]=\left[\frac{E}{L}\right]e^{-t/\lambda}$$

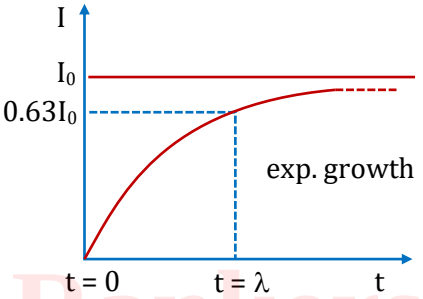
$$t=0,\left[-\frac{di}{dt}\right]_{max.}=\frac{E}{L}$$

$$t\to\infty\Rightarrow\left[-\frac{di}{dt}\right]\to 0$$

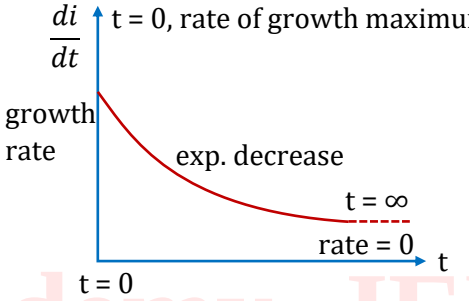
Graph for R-L circuit:

Current Growth:

(a)

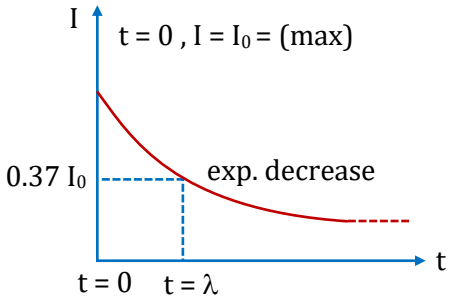


(b)

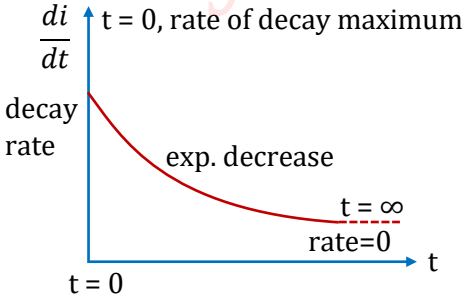


Current decay:

(a)



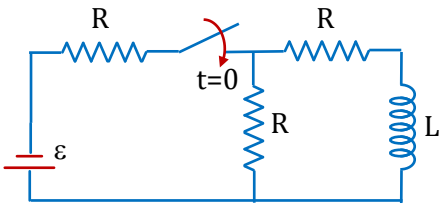
(b)



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Illustration 27

In the following circuit the switch is closed at $t = 0$. Initially there is no current in inductor. Find out current in the inductor coil as a function of time.



Solution:

At any time t

$$-\varepsilon+i_1R-(i-i_1)R=0$$

$$-\varepsilon+2i_1R-iR=0$$

$$i_1=\frac{iR+\varepsilon}{2R}$$

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Now, $-\varepsilon + i_1 R + iR + L.\frac{di}{dt} = 0 \quad -\varepsilon + \left(\frac{iR+\varepsilon}{2}\right) + iR + L.\frac{di}{dt} = 0$

$$-\frac{\varepsilon}{2} + \frac{3iR}{2} = -L.\frac{di}{dt}$$

$$\left(\frac{-\varepsilon + 3iR}{2}\right)dt = -L.di$$

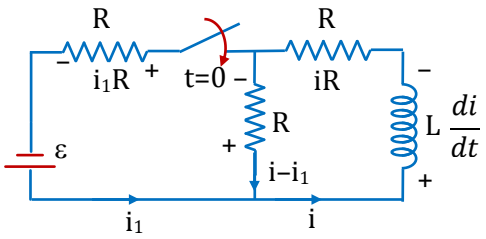
$$\frac{-dt}{2L} = \frac{di}{-\varepsilon + 3iR}$$

$$-\int_0^t \frac{dt}{2L} = \int_0^i \frac{di}{-\varepsilon + 3iR}$$

$$-\frac{t}{2L} = \frac{1}{3R}\ln\left(\frac{-\varepsilon + 3iR}{-\varepsilon}\right)$$

$$-\ln\left(\frac{-\varepsilon + 3iR}{-\varepsilon}\right) = \frac{3Rt}{2L}$$

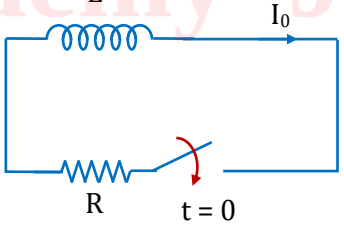
$$i = +\frac{\varepsilon}{3R}\left(1 - e^{-\frac{3Rt}{2L}}\right)$$



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Illustration 28

Prove that decay energy loss in resistor from $t = 0$ to ∞ is equal to $\frac{1}{2}Li_0^2$.



Solution:

$$dH = \int i^2 r dt$$

$$\Rightarrow \quad dH = \int \left(i_o e^{\frac{-t}{L/R}}\right)^2 R dt$$

$$H\Big|_0^H = \int i_0^2 R \left(e^{\frac{-tR}{L}}\right)^2 dt$$

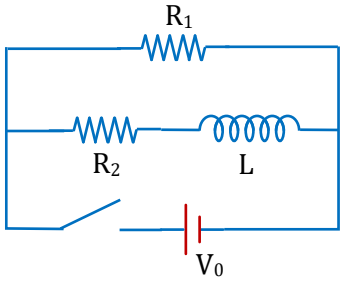
$$H - 0 = i_0^2 R \int e^{\frac{-2tR}{L}} dt$$

$$H - 0 = i_0^2 R \int e^{\frac{-2tR}{L}} dt = i_0^2 R e^{\frac{-2tR}{L}} \left(\frac{-L}{2R}\right)_0^\infty = \frac{1}{2}Li_0^2$$

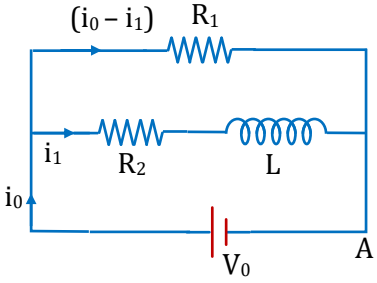
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Illustration 29

Switch was open for long time and now it is closed at $t = 0$. Find current from R_2 as a function of time.



Solution:



$$V_A + V_0 - i_1R_2 - \frac{Ldi_1}{dt} = V_A$$

$$V_0 - i_1R_2 = \frac{Ldi_1}{dt}$$

$$\frac{-dt}{L} = \frac{di}{(i_1R_2 - V_0)} \quad \Rightarrow \quad \frac{-t}{L}\Big|_0^t = \int_0^{i_1} \frac{\ell n(i_1R_2 - V_0)}{R_2}$$

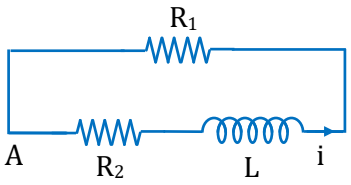
$$\frac{-t}{L} = \ell n\left(\frac{i_1R_2 - V_0}{-V_0}\right) \times \frac{1}{R_2} \quad \Rightarrow \quad \frac{-tR_2}{L} = \ell n\left(\frac{i_1R_2 - V_0}{-V_0}\right)$$

$$\frac{-tR_2}{L} = \ell n\left(\frac{V_0 - i_1R_2}{V_0}\right)$$

$$\frac{V_0 - i_1R_2}{V_0} = e^{\frac{-tR_2}{L}} \quad \Rightarrow \quad V_0 - i_1R_2 = V_0e^{\frac{-tR_2}{L}}$$

$$i_1R_2 = V_0\left(1 - e^{-tR_2/L}\right) \quad \Rightarrow \quad i_1 = \frac{V_0}{R_2}\left(1 - e^{-tR_2/L}\right)$$

$$i_1 = \frac{V_0}{R_2}\left(1 - e^{-tR_2/L}\right) \quad \Rightarrow \quad i_0 = \frac{V_0}{R_2}$$



$$V_A - iR_2 - \frac{Ldi}{dt} - iR_1 = V_A$$

$$\frac{Ldi}{dt} = -i(R_1 + R_2) \quad \Rightarrow \quad \frac{di}{-i(R_1 + R_2)} = \frac{dt}{L}$$

$$-\frac{di}{i} = \left(\frac{R_1 + R_2}{L}\right)dt \quad \Rightarrow \quad \ell ni\Big|_{i_0}^{i} = \left|\left(\frac{R_1 + R_2}{L}\right)t\right|_0^t$$

$$-\ell n\frac{i}{i_0} = \left(\frac{R_1 + R_2}{L}\right)t \quad \Rightarrow \quad i = i_0e^{\frac{-t}{L}(R_1 + R_2)}$$

$$i_0 = \frac{V_0}{R_2}$$

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Energy stored in an inductor

The energy of a capacitor is stored in the electric field between its plates. Similarly, an inductor has the capability of storing energy in its magnetic field. An increasing current in an inductor causes an emf between its terminals.

Power P = The work done per unit time $= \frac{dW}{dt} = -ei = -\left[L\frac{di}{dt}\right]i = -Li\frac{di}{dt}$

Here i = instantaneous current and L=inductance of the coil

From $dW = -dU$ (energy stored) so $= \frac{dW}{dt} = -\frac{dU}{dt}$ $\quad\therefore\quad \frac{dU}{dt} = Li\frac{di}{dt} \Rightarrow dU = Li\,di$

The total energy U supplied while the current increases from zero to final value I is.

$$U = L\int_0^I i\,di = \frac{1}{2}L\left(i^2\right)_0^I \quad \therefore\ U = \frac{1}{2}LI^2$$

the energy stored in the magnetic field of an inductor when a current I is $= \frac{1}{2}LI^2$.

The source of this energy is the external source of emf that supplies the current.

After the current has reached its final steady state value I, $\frac{di}{dt} = 0$ and no more energy is input to the inductor.

When the current decreases from I to zero, the inductor acts as a source that supplies a total amount of energy $\frac{1}{2}Li^2$ to the external circuit. If we interrupt the circuit suddenly by opening a switch the current decreases very rapidly, the induced emf is very large and the energy may be dissipated.

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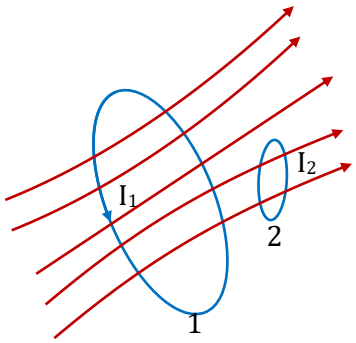
Magnetic energy density = $\frac{1}{2}\frac{B^2}{\mu_0}$

Mutual inductance

Consider two arbitrary conducting loops 1 and 2. Suppose that I_1 is the instantaneous current flowing around loop 1. This current generates a magnetic field B_1 which links the second circuit, giving rise to a magnetic flux ϕ_2 through that circuit. If the current I_1 doubles, then the magnetic field B_1 doubles in strength at all points in space, so the magnetic flux ϕ_2 through the second circuit also doubles. Furthermore, it is obvious that the flux through the second circuit is zero whenever the current flowing around the first circuit is zero. It follows that the flux ϕ_2 through the second circuit is directly proportional to the current I_1 flowing around the first circuit. Hence, we can write $\phi_2 = M_{21}I_1$ where the constant of proportionality M_{21} is called the mutual inductance of circuit 2 with respect to circuit 1. Similarly, the flux ϕ_2 through the first circuit due to the instantaneous current I_2 flowing around the second circuit is directly proportional to that current, so we can write $\phi_1 = M_{12}I_2$ where M_{12} is the mutual inductance of circuit 1 with respect to circuit 2. It can be shown that $\boldsymbol{M_{21} = M_{12}}$ **(Reciprocity Theorem)**. Note that M is a purely geometric quantity, depending only on the size, number of turns, relative position, and relative orientation of the two circuits. The S.I. unit of mutual inductance is called Henry (H). One Henry is equivalent to a volt-second per ampere.

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Suppose that the current flowing around circuit 1 changes by an amount ΔI_1 in a small time interval Δt. The flux linking circuit 2 changes by an amount $\Delta\phi_2 = M\Delta I_1$ in the same time interval. According to Faraday's law, an emf $\varepsilon_2 = -\frac{\Delta\phi_2}{\Delta t}$ is generated around the second circuit due to the changing magnetic flux linking that circuit. Since, $\Delta\phi_2 = M\Delta I_1$, this emf can also be written $\varepsilon_2 = -M\frac{\Delta I_1}{\Delta t}$.



Thus, the emf generated around the second circuit due to the current flowing around the first circuit is directly proportional to the rate at which that current changes. Likewise, if the current I_2 flowing around the second circuit changes by an amount ΔI_2 in a time interval Δt then the emf generated around the first circuit is $\varepsilon_1 = -M\frac{\Delta I_2}{\Delta t}$ Note that there is no direct physical connection(coupling) between the two circuits: the coupling is due entirely to the magnetic field generated by the currents flowing around the circuits.

Note:

(1) $M \le \sqrt{L_1 L_2}$

(2) For two coils in series if mutual inductance is considered then

$L_{eq} = L_1 + L_2 \pm 2M$

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Illustration 30

Two insulated wires are wound on the same hollow cylinder, so as to form two solenoids sharing a common air-filled core. Let I be the length of the core, A the cross-sectional area of the core, N_1 the number of times the first wire is wound around the core, and N_2 the number of turns the second wire is wound around the core. Find the mutual inductance of the two solenoids, neglecting the end effects.

Solution:

If a current I_1 flows around the first wire then a uniform axial magnetic field of strength $B_1 = \frac{\mu_0 N_1 I_1}{\ell}$ is generated in the core. The magnetic field in the region outside the core is of negligible magnitude. The flux linking a single turn of the second wire is $B_1 A$. Thus, the flux linking all N_2 turns of the second wire is

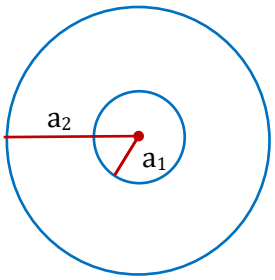
$$\phi_2 = N_2 B_1 A = \frac{\mu_0 N_1 N_2 I_1 A}{\ell} = MI_1 \qquad \therefore \quad M = \frac{\mu_0 N_1 N_2 A}{\ell}$$

As described previously, M is a geometric quantity depending on the dimensions of the core and the manner in which the two wires are wound around the core, but not on the actual currents flowing through the wires.

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Illustration 31

Find the mutual inductance of two concentric coils of radii a_1 and a_2 ($a_1 << a_2$) if the planes of coils are same.



Solution:

Let a current i flow in coil of radius a_2.

Magnetic field at the centre of coil = $\frac{\mu_0 i}{2a}$

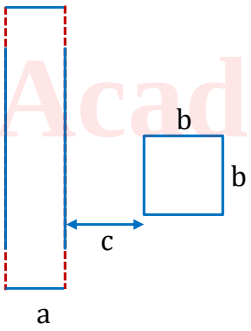
Flux linkage = Mi

$Mi = \frac{\mu_0 i}{2a}\pi a_1^2$ or $M = \frac{\mu_0 \pi a_1^2}{2a_2}$

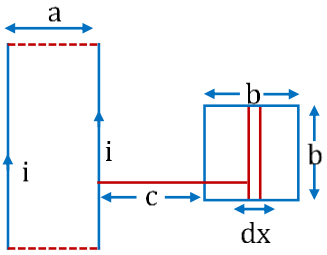
Illustration 32

Find the mutual inductance between two rectangular loops, shown in the figure.

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Solution:



Let current i flow in the loop having ∞-ly long sides. Consider a segment of width dx at a distance x as shown flux through the region

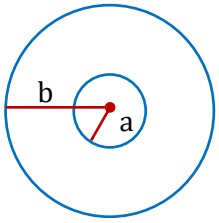
$$d\phi = \left[\frac{\mu_0 i}{2\pi x} - \frac{\mu_0 i}{2\pi(x+a)}\right] b\,dx$$

$$\Rightarrow \quad \phi = \int_c^{c+b}\left[\frac{\mu_0 i}{2\pi x} - \frac{\mu_0 i}{2\pi(x+a)}\right] b\,dx = \frac{\mu_0 i b}{2\pi}\left[\ell n\frac{c+b}{c} - \ell n\frac{a+b+c}{a+c}\right] = Mi$$

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Illustration 33

Figure shows two concentric coplanar coils with radii a and b ($a << b$). A current $i = 2t$ flows in the smaller loop. Neglecting self inductance of larger loop.



(a) Find the mutual inductance of the two coils.

(b) Find the emf induced in the larger coil.

(c) If the resistance of the larger loop is R find the current in it as a function of time.

Solution:

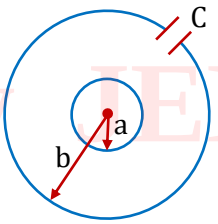
(a) To find mutual inductance, it does not matter in which coil we consider current and in which flux is calculated (Reciprocity theorem) Let current i be flowing in the larger coil. Magnetic field at the centre = $\frac{\mu_0 i}{2b}$.

Flux through the smaller coil = $\frac{\mu_0 i}{2b}\pi a^2$ $\quad\therefore\quad M = \frac{\mu_0}{2b}\pi a^2$

(b) $\left|\text{emf induced in larger coil}\right| = M\left[\left(\frac{di}{dt}\right)\text{in smaller coil}\right] = \frac{\mu_0}{2b}\pi a^2(2) = \frac{\mu_0 \pi a^2}{b}$

(c) Current in the larger coil = $\frac{\mu_0 \pi a^2}{bR}$.

Illustration 34



If the current in the inner loop changes according to $i = 2t^2$ then, find the current in the capacitor as a function of time. ($a << b$)

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Solution:

$$M = \frac{\mu_0}{2b}\pi a^2$$

$$\left|\text{emf induced in larger coil}\right| = M\left[\left(\frac{di}{dt}\right)\text{in smaller coil}\right]$$

$$\Rightarrow \quad e = \frac{\mu_0}{2b}\pi a^2(4t) = \frac{2\mu_0 \pi a^2 t}{b}$$

Applying KVL :

$$+e - \frac{q}{C} - iR = 0$$

$$\frac{2\mu_0 \pi a^2 t}{b} - \frac{q}{C} - iR = 0$$

Differentiate wrt time :-

$$\frac{2\mu_0 \pi a^2}{b} - \frac{i}{C} - \frac{di}{dt}R = 0$$

On solving it

$$i = \frac{2\mu_0 \pi a^2 C}{b}\left[1 - e^{-t/RC}\right]$$

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Coupled Coils:

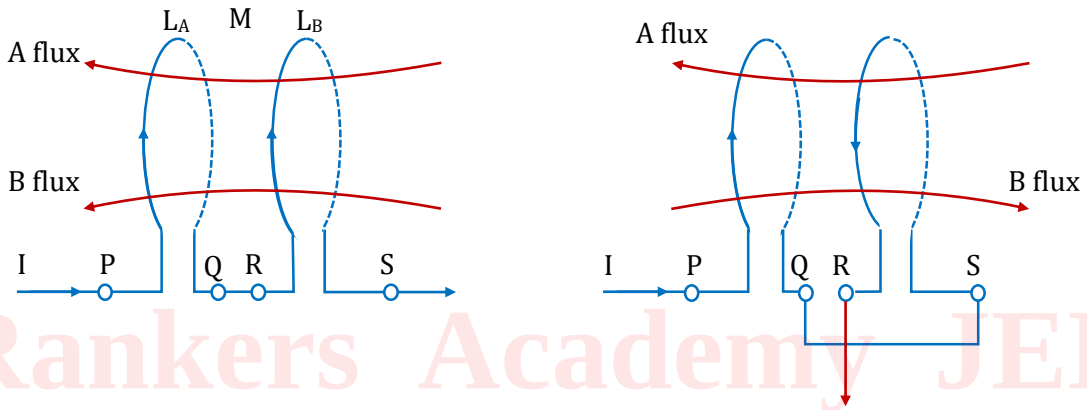
The coils of figure have a mutual inductance of magnitude M. If Q and R are connected and a current I passed through both coils in series then, considered as one circuit, the total flux linkage is $L_A I + L_B I + 2MI$ and the

$$L_1 = L_A + L_B + 2M$$

If, on the other hand, Q and S are connected and a current passed from P to R the total flux linkage is now $L_A I + L_B I + 2MI$ (the self-fluxes must be positive and the mutual fluxes oppose them) so that

$$L_2 = L_A + L_B - 2M$$

The same effect could be obtained by retaining the connections but rotating one coil through 180°–*a* method which can therefore be used to obtain a self inductance varying smoothly from L_1 to L_2. Alternatively, because $M = \frac{1}{4}(L_1 - L_2)$ from above equation a mutual inductance may be obtained from measurements only of self-inductance, which are often simpler.



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Figure : Two coupled coils.

Magnetic energy due to mutual inductance

Consider now two circuits with self inductances L_1 and L_2 and a mutual inductance M_{21}. We do not now assume that $M_{21} = M_{12}$.

First insert an electromotance (e.g. a cell) in circuit 1 so that the current in it grows as above to its final value I_1, keeping any induced currents in 2 at zero by extra electromotances (these do no work since $I_2 = 0$). The energy stored, as above, is $\frac{1}{2}L_1 I_1^2$.

Now insert an electromotance in circuit 2 to make the current in it grow to its final value I_2. When it has done so, naturally an energy $\frac{1}{2}L_2 I_2^2$ is stored in 2, but while I_2 is changing an extra electromotance $-M_{21}di_2/dt$ is induced in circuit 1, i_2 being the instantaneous value. We insert in 1 an adjustable electromotance of value $+M_{21}di_2/dt$, thus keeping I_1 constant (and not upsetting the $\frac{1}{2}L_1 I_1^2$ already stored). This time, however, the extra electromotance passes the current I_1 and thus puts energy into the system at the rate $I_1 M_{21} di_2/dt$. The total energy put in by this extra electromotance is therefore $I_1\, M_{21} di_2$ integrated from 0 to I_2, or $M_{21} I_1 I_2$.

Hence for a pair of circuits

$$U_M = \frac{1}{2}L_1 I_1^2 + M_{21} I_1 I_2 + \frac{1}{2}L_2 I_2^2$$

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And because the same situation can be reached by allowing I_2 to grow first and finding that U_M is then $\frac{1}{2}L_1I_1^2 + M_{21}I_1I_2 + \frac{1}{2}L_2I_2^2$, it follows that $M_{12} = M_{21}$

There is only one mutual inductance M between two circuits and the magnetic energy is

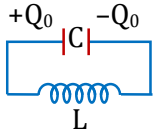
Two coupled circuits: $\boxed{U_M = \frac{1}{2}L_1I_1^2 + M_{21}I_1I_2 + \frac{1}{2}L_2I_2^2}$

This can also be written as

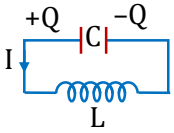
$$U_M = \frac{1}{2}I_1(L_1I_1 + MI_2) + \frac{1}{2}I_2(L_2I_2 + MI_1) = \frac{1}{2}I_1\phi_1 + \frac{1}{2}I_2\phi_2$$

Where ϕ_1 and ϕ_2 are total magnetic fluxes linking each circuit.

LC oscillations:



At $t = 0$



At $t = t$

When capacitor C is completely charged up to Q_0 and connected to an inductor L at $t = 0$ then at $t = t$

$$L\frac{dI}{dt} - \frac{Q}{C} = 0, \qquad -L\frac{d^2Q}{dt^2} - \frac{Q}{C} = 0 \quad Q = -LC\frac{d^2Q}{dt^2}$$

$\frac{d^2Q}{dt^2} + \frac{1}{LC}Q = 0$ therefore, charge Q oscillates with $Q = Q_0 \cos \omega t$

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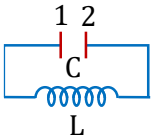
Hence initial phase of oscillation is $\frac{\pi}{2}$ and angular frequency $\omega = \frac{1}{\sqrt{LC}}$

One can prove that the energy in the system remains conserved.

Therefore $\frac{Q_0^2}{2C} + 0 = \frac{Q^2}{2C} + \frac{1}{2}LI^2 = \frac{1}{2}LI_0^2 + 0$

Illustration 35

Consider a $L - C$ oscillation circuit. Circuit elements has zero resistance. Initially at $t = 0$ all the energy is stored in the form of electric field and plate-1 is having positive charge:



At time $t = t_1$ plate-2 attains half of the maximum *+ve* change for the first time. Value of t_1 is :

(A) $\frac{2\pi}{3}\sqrt{LC}$ (B) $\frac{\pi}{3}\sqrt{LC}$

(C) $\frac{4\pi}{3}\sqrt{LC}$ (D) $\pi\sqrt{LC}$

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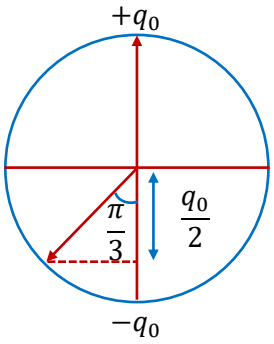
Solution:

$q_1 = q_0 \sin(\omega t + \pi/2)$

At $t = t_1, \quad q_1 = -\frac{q_0}{2}$

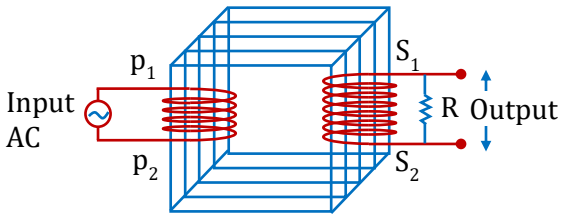
$$t_1 = \frac{\pi - \frac{\pi}{3}}{\omega}$$

$$= \frac{2\pi}{3\omega} = \frac{2\pi}{3}\sqrt{LC}$$



Transformers

One of the great advantages of ac over dc for electric-power distribution is that it is much easier to step voltage levels up and down with ac than with dc. The necessary conversion is accomplished by a static device called transformer using the principle of mutual induction.



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The figure shows an idealised transformer which consists of two coils or windings, electrically insulated from each other but wound on the same core. The winding to which power is supplied is called primary, the winding from which power is delivered is called the secondary.

The ac source causes an alternating current in the primary which sets up an alternating flux in the core and this induces an emf in each winding of secondary in accordance with Faraday's law. For ideal transformer we assume that primary has negligible resistance and all the flux in core links both primary and secondary. The primary winding has N_1 turns and secondary has N_2 turns. When the magnetic flux changes because of changing currents in the two coils, the resulting induced emf are

$$E_1 = -N_1 \frac{d\phi_B}{dt} \text{ and } E_2 = -N_2 \frac{d\phi_B}{dt}$$

The flux per turn B is same in both primary and the secondary so that the emf per turn is same in each. The ratio of secondary emf E_2 to the primary emf E_1 is therefore equal at any instant to the ratio of secondary to primary turns.

$$\frac{E_2}{E_l} = \frac{N_2}{N_1}$$

If the windings have zero resistance, the induced emf 1 and 2 are equal to the terminal voltage across the primary and the secondary respectively, hence

$$\frac{V_2}{V_l} = \frac{N_2}{N_1}$$

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If $N_2 > N_1$ then $V_2 > V_1$ and we have step up transformer, if $N_2 < N_1$ then $V_2 < V_1$ and we have a step down transformer.

If the transformer is assumed to be 100% efficient (no energy losses) the power input is equal to the power output i.e.

$$I_2V_2 = I_1V_1$$

$$\therefore \quad \frac{I_2}{I_1} = \frac{N_1}{N_2}$$

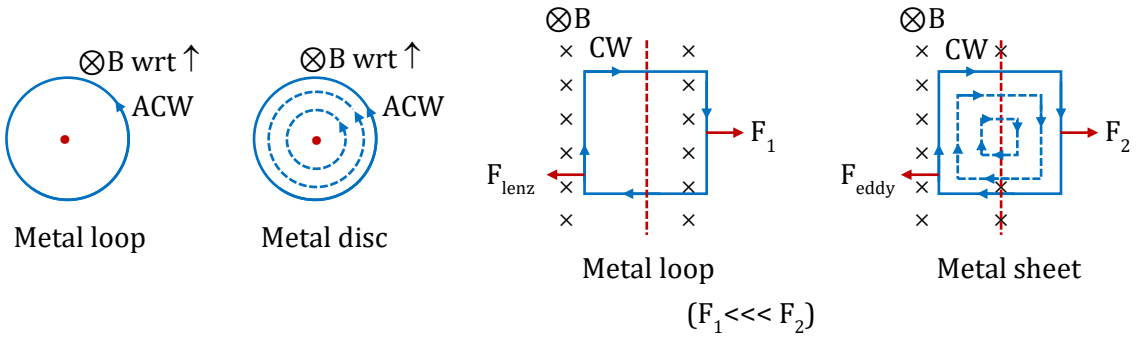
All the currents and voltages derived above have same frequency as that of source. The equations obtained above apply to ideal transformers, although some energy is lost but well designed transformers have efficiency more than 95%, this is a good approximation. The causes of energy losses and their rectification is given below

	Cause		Rectification
1.	Due to poor design and air gaps in the core, all the flux due to primary does not pass through the secondary.	**1.**	By winding the primary and secondary coil one over the other.
2.	Resistance of windings causes I^2R loss.	**2.**	In high current, low voltage, these are minimised by using thick wire.
3.	The alternating magnetic flux induces eddy currents in the core and causes heating.	**3.**	By using laminated core it can be reduced.
4.	Alternating magnetisating of core causes hystersis loss.	**4.**	It is kept minimum by using a magnetic material having low hystersis loss. (e.g. soft iron)

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Eddy current :

Eddy currents (or Focalt's currents) [Experimental verification by Focalt]



Def. It is a group of induced currents which are produced, when metal bodies placed in time varying magnetic field or they moves in external magnetic field in such a way that flux through them changes with respect to time.

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